CAD for VLSI

Course Outline

CS5140 CAD

- Instructor:
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- EDA Where electronics begins: YouTube Clip

Reference Text Book

- Synthesis and Optimization of Digital Circuits, Giovanni De Micheli, 1st Edition

- Electronic Design Automation: Synthesis, Verification, and Test

  L. T. Wang, Y. W. Chang, K. T. Cheng, 1st Edition

YouTube Playlists for your reference

https://www.youtube.com/c/YiYuLiu/playlists

Course Objectives

- Obtain a general understanding of IC designs
- Understand the circuit modeling
- Study the basic algorithms used in architectural synthesis
- Learn about the high level synthesis, logic synthesis, and physical design automation techniques used in the best-known academic and commercial synthesis systems
- Get know hot research topics and problems
- Sharpen your C/C++ programming skill

Intended Audience

- VLSI CAD (also known as EDA – electronic design automation) students
- Circuit designers to understand how tools work behind the scene
- Architect to optimize design among different objectives, area, power, performance, and etc.
- Mathematical/Computer Science majors who want to find tough problems to solve
  - Lots of VLSI synthesis problems can be formulated into combinatorial optimization or mathematical programming problems.
  - Actually, most CAD problems are NP-completeness problems → we need heuristics

Course Outline

- Introduction
- Modeling
- Scheduling
- Resource binding
- Boolean representation
- Two-level logic optimization
- Multi-level logic optimization
- Technology mapping
- Partitioning
- Floorplanning
- Placement
- Global routing
- Detailed routing
- Clock and power routings
- Advanced packaging and emerging topics

Grading

- Homework assignment: 75% - 80%
  - Kick-off project * 1
  - Programming assignments * 3
    - Performance 80%
    - Algorithm 10%
    - Research & Optimization 10%
  - Final research project * 1
    - Team of 2~3
- In-class quizzes: 20%
- Class interaction: 0% - 5%


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